

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 2

3 June 2021

Time : 3 hours

Maximum marks: 52

Full marks: 45

- This question paper consists of 4 pages (including this one).
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

Section A

In this section, give your final answer only. Number the answers clearly. Do not show your working.

Question 1. [3 points] Let $V = \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}, x_1 > 0\}$. Define, for $(x_1, x_2), (y_1, y_2) \in V, \lambda \in \mathbb{R}$,

$$\begin{aligned}(x_1, x_2) \oplus (y_1, y_2) &= (x_1 y_1, x_2 + y_2) \\ \lambda \odot (x_1, x_2) &= (x_1^\lambda, \lambda x_2)\end{aligned}$$

Then it can be shown that V with addition $\oplus$ and scalar multiplication $\odot$ is a vector space.

- (a) Find $((-3) \odot (2, -5)) \oplus (4, -3)$.
- (b) Find the additive identity (zero element) of V .
- (c) Find the additive inverse of (x_1, x_2) .

Question 2. [7 points] The following is a list of subsets of $M_{2 \times 2}(\mathbb{R})$:

$$\begin{aligned}S_1 &= \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \\ S_2 &= \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \\ S_3 &= \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \\ S_4 &= \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}\end{aligned}$$

Choose, if possible, a set as described in (a) to (d) below. For your answer, write S_1, S_2, S_3, S_4 or "Impossible". You may use a set more than once, or not at all.

- (a) A set S_i which is a basis for $M_{2 \times 2}(\mathbb{R})$.
- (b) A set S_i such that $\dim(\text{span}(S_i)) = 2$.
- (c) A linearly dependent set S_i which spans $M_{2 \times 2}(\mathbb{R})$.
- (d) A set S_i such that the set obtained by removing $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ from S_i is a basis for $\text{span}(S_i)$.

Question 3. [7 points] Let

$$A = \begin{pmatrix} 1 & -1 & 2 & 3 & 4 \\ 3 & -1 & 3 & 9 & 13 \\ 1 & -3 & 5 & 3 & 3 \\ 2 & 0 & 1 & 6 & 8 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -1 & 2 & 3 & 4 & a \\ 3 & -1 & 3 & 9 & 13 & b \\ 1 & -3 & 5 & 3 & 3 & c \\ 2 & 0 & 1 & 6 & 8 & d \end{pmatrix},$$

$$U = \begin{pmatrix} 1 & -1 & 2 & 3 & 4 & a \\ 0 & 2 & -3 & 0 & 1 & b - 3a \\ 0 & 0 & 0 & 0 & 1 & 3a - c - d \\ 0 & 0 & 0 & 0 & 0 & 4a - b - c \end{pmatrix}.$$

The matrix U was obtained from the matrix C by performing elementary row operations on C . **Use this information** to answer the following questions:

- (a) What is the nullity and rank of A ?
- (b) What is the nullity of A^T ? (A^T is the transpose of A .)
- (c) Find a basis for $NS(A)$.
- (d) Write down a basis for $CS(A)$.
- (e) Write down a basis for $RS(A)$.
- (f) An incomplete description of $CS(A)$ is given below; there should be an equation in the place of the dots

$$CS(A) = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \in \mathbb{R}^4 : \dots \right\}$$

Write down this equation.

Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [4 points] Let A be the matrix in Question 3.

- (a) Let B be a basis for $CS(A)$. Find, if possible, a vector $\mathbf{v}$ such that $B \cup \{\mathbf{v}\}$ is a basis for $\mathbb{R}^4$, and explain why $B \cup \{\mathbf{v}\}$ is a basis for $\mathbb{R}^4$. If it is not possible, explain why it is not possible.
- (b) Is it possible to obtain an invertible matrix B by deleting a column of A ? If it is, say which column should be deleted, and why the resulting matrix is invertible. If it is not, explain why it is not possible.

Question 5. [10 points] Let $S = \{p(x) \in P_3(x) : p(0) = 0 \text{ and } p(1) \geq 1\}$. Answer the following questions, giving reasons for your answers.

- (a) Is S closed under addition?
- (b) Is S a subspace of $P_3(x)$?
- (c) Is $R = \{x^3 - 3x^2 - 6x, 2x^3 - 3x, 4x^2 + 8x\}$ a linearly independent subset of $P_3(x)$?
- (d) Find $\dim(\text{span}(R))$.

Question 6. [8 points] Let V be the vector space of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ and let the function $T : V \rightarrow V$ be defined by

$$T(f) = g, \text{ where } g \text{ is defined by } g(x) = f(0) + f'(0)x, \text{ for every } x \in \mathbb{R}.$$

- (a) Show that T is a linear transformation.
- (b) Find the range of T and its dimension. Give reasons for your answers.
- (c) Give an example of a non-zero function in $\ker(T)$.
- (d) Is $\ker(T)$ finite dimensional? Give a reason for your answer.

Question 7. [8 points] Read the following proof:

Proof. (I) Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent subset of a vector space V and let $\mathbf{v} \in V$ be such that $\mathbf{v} \notin \text{span}(S)$. Suppose there are scalars $a_1, a_2, \dots, a_n, a_{n+1}$, not all zero, such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n + a_{n+1}\mathbf{v} = \mathbf{0}. \quad (1)$$

Then we cannot have $a_{n+1} = 0$, since this would contradict the linear independence of S . (2)

But if $a_{n+1} \neq 0$, then

$$\mathbf{v} = -\frac{1}{a_{n+1}}[a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n]. \quad (3)$$

But this contradicts our assumption about $\mathbf{v}$. It follows that $S \cup \{\mathbf{v}\}$ must be linearly independent.

(II) Let S be as in (a) and suppose $\dim(V) = n$. If there is a $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span}(S)$, then $S \cup \{\mathbf{v}\}$ is linearly independent. But this contradicts the fact that $\dim(V) = n$. (4)

Hence S is a basis for V . (5) $\square$

Now answer the following questions:

- (a) State the results proved in **(I)** in full.
- (b) What is assumed about the set $S \cup \{\mathbf{v}\}$ at (1)?
- (c) Explain how, at (2), $a_{n+1} = 0$ would contradict the linear independence of S .
- (d) Which assumption about $\mathbf{v}$ is contradicted by the equation (3)?
- (e) State the results proved in **(II)** in full.
- (f) Explain what the contradiction is that arises at (4).
- (g) Why can we claim at (5) that S is a basis for V ?

Question 8. [5 points] Prove or disprove:

- (a) If C and D are square matrices of the same size and their column spaces are equal, then their null spaces are also equal.
- (b) If V is a finite-dimensional vector space and W is a subspace of V such that $\dim(W) = \dim(V)$, then $W = V$.
- (c) If V is an infinite-dimensional vector space and W is an infinite-dimensional subspace of V , then $W = V$.